线性回归

bedroom cnt	Size	Price
3	2104	400
2	1416	232
3	1334	315
2	852	178

假设函数

$h(x) = a + bx_1 + cx_2$

令 $h(x) = \sum_{j=0}^{2} \theta_j x_j$, $x_0 = 1$

$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$, $X = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$, $Y = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix}$

M为样本数，θ为参数，X为输入或特征，Y为输出或目标变量

(x,y)为训练样本，$(x^{(i)}, y^{(i)})$为第i个训练样本，n为特征数量

最小化 $\underbrace{\frac{1}{2}\sum_{i=1}^{m}\left[h_\theta(x^{(i)}) - y^{(i)}\right]^2}_{J(\theta)}$，其中$\theta$为变量

梯度下降

选取一个初值θ，再不断减少$J(\theta)$，每次选择一个尽可能快的方向下坡

尽能会得到局部最优解，在线性回归中其是全局最优解

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

$$\frac{\partial}{\partial \theta_j} J(\theta) = \sum_{i=1}^{m} \frac{\partial}{\partial \theta_j} \frac{1}{2}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 = \sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right) \cdot \frac{\partial}{\partial \theta_j}\left(h_\theta(x^{(i)}) - y^{(i)}\right)$$

$$= \sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_j^{(i)}$$

$$\theta_j := \theta_j - \alpha \sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right) x_j^{(i)}$$

批量梯度下降：将数据看成一个整体一起处理，过慢。每次用所有训练集更新

随机梯度下降：每次随机一个训练集i，去更新θ。

无论如何，都要整体更新θ，而不能先更新θ_1，再更新θ_2。

因为θ_1的更新导致θ变了，那θ_2梯度计算也会变化。

也即要动，就所有维度一起动，而不能只动某一维度。

本质是每次迭代选取一个训练子集，即i可分离。再尽可能将

θ去拟合选取的训练子集。但θ更新是一步走，即θ是整体的。j不可分离。

$$\nabla_\theta J(\theta) = \begin{bmatrix} \frac{\partial J}{\partial \theta_0} \\ \vdots \\ \frac{\partial J}{\partial \theta_n} \end{bmatrix}$$

$\theta \in \mathbb{R}^{n+1}$

$A \in \mathbb{R}^{2\times 2}$, $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$f(A) = A_{11} + A_{12}^2$ $\quad f: \mathbb{R}^{2\times 2} \to \mathbb{R}$

$$\nabla_A f(A) = \begin{bmatrix} \frac{\partial f}{\partial a_{11}} & \frac{\partial f}{\partial a_{12}} \\ \frac{\partial f}{a_{21}} & \frac{\partial f}{a_{22}} \end{bmatrix} = \begin{bmatrix} 1 & 2a_{12} \\ 0 & 0 \end{bmatrix}$$

$A \in R^{m\times n}$, $f: R^{m\times n} \to R$

则 $\nabla_A f(A) = \begin{bmatrix} \frac{\partial f}{\partial a_{11}} & \cdots & \frac{\partial f}{\partial a_{1n}} \\ \vdots & \ddots & \\ \frac{\partial f}{\partial a_{m1}} & \cdots & \frac{\partial f}{\partial a_{mn}} \end{bmatrix}$

例 $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$f(A) = \frac{3}{2}a_{11} + 5a_{12}^2 + 2a_{21}a_{22}$

$\nabla_A f(A) = \begin{bmatrix} \frac{3}{2} & 10a_{12} \\ 2a_{22} & 2a_{21} \end{bmatrix}$

$A \in R^{n\times n}$，则 $trA = \sum_{i=1}^{n} a_{ii}$，称为矩阵的迹

迹的一些性质：

$trA_1 \dots A_n = trA_nA_1 \dots A_{n-1}$

$trA = trA^T$，$tr(A+B) = trA + trB$，$traA = atrA$

$\nabla_A trAB = B^T$

$\nabla_{A^T} f(A) = (\nabla_A f(A))^T$

$\nabla_A trABA^TC = CAB + C^TAB^T$

$\nabla_A |A| = |A|(A^{-1})^T$

令 $X = \begin{bmatrix} (x^{(1)})^T \\ \vdots \\ (x^{(m)})^T \end{bmatrix}$，$Y = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(m)} \end{bmatrix}$，$X\theta - Y = \begin{bmatrix} h_\theta(x^{(1)}) - y^{(1)} \\ \vdots \\ h_\theta(x^{(m)}) - y^{(m)} \end{bmatrix}$

则有 $J(\theta) = \frac{1}{2}(X\theta - Y)^T(X\theta - Y)$

$$\begin{aligned}
\nabla_\theta J(\theta) &= \frac{1}{2}\nabla_\theta(\theta^TX^T - Y^T)(X\theta - Y) \\
&= \frac{1}{2}\nabla_\theta(\theta^TX^TX\theta - \theta^TX^TY - Y^TX\theta + Y^TY) \\
&= \frac{1}{2}\nabla_\theta tr(\theta^TX^TX\theta - \theta^TX^TY - Y^TX\theta + Y^TY) \\
&= \frac{1}{2}\nabla_\theta(tr\,\theta^TX^TX\theta - 2tr\,Y^TX\theta) \\
&= \frac{1}{2}\nabla_\theta(X^TX\theta + X^TX\theta - 2X^TY) \\
&= X^TX\theta - X^TY = 0
\end{aligned}$$

则 $X^TX\theta = X^TY$，$\theta = (X^TX)^{-1}X^TY$

令 Y' 为 Y 在 X 列空间中投影，则 $Y' = X(X^TX)^{-1}X^TY$

即求解 $X\theta = Y' = X(X^TX)^{-1}X^TY$

也得 $\theta = (X^TX)^{-1}XY$